

Integrating Individual and Group Fairness for Recommender Systems through Social Choice

Amanda Aird
University of Colorado Boulder
Boulder, USA
amanda.aird@colorado.edu

Elena Štefancová
Comenius University Bratislava
Bratislava, Slovakia
elenastefancova@gmail.com

Anas Buhayh
University of Colorado Boulder
Boulder, USA
Anas.Buhayh@colorado.edu

Cassidy All
Department of Information Science;
University of Colorado, Boulder
Boulder, USA
cassidy.all@colorado.edu

Martin Homola
Comenius University Bratislava
Bratislava, Slovakia
homola@fmph.uniba.sk

Nicholas Mattei
Tulane University
New Orleans, USA
nsmattei@gmail.com

Robin Burke
University of Colorado, Boulder
Boulder, USA
robin.burke@colorado.edu

Abstract

Fairness in recommender systems is a complex concept, involving multiple definitions, different parties for whom fairness is sought, and various scopes over which fairness might be measured. Researchers seeking fairness-aware systems have derived a variety of solutions, usually highly tailored to specific choices along each of these dimensions, and typically aimed at tackling a single fairness concern, i.e., a single definition for a specific stakeholder group and measurement scope. However, in practical contexts, there are a multiplicity of fairness concerns within a given recommendation application and solutions limited to a single dimension are therefore less useful. We explore a general solution to recommender system fairness using social choice methods to integrate multiple heterogeneous definitions. In this paper, we extend group-fairness results from prior research to provider-side individual fairness, demonstrating in multiple datasets that both individual and group fairness objectives can be integrated and optimized jointly. We identify both synergies and tensions among different objectives with individual fairness correlated with group fairness for some groups and anti-correlated with others.

Keywords

recommender systems, fairness, computational social choice

ACM Reference Format:
Amanda Aird, Elena Štefancová, Anas Buhayh, Cassidy All, Martin Homola, Nicholas Mattei, and Robin Burke. 2025. Integrating Individual and Group Fairness for Recommender Systems through Social Choice. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25), September 22–26, 2025, Prague, Czech Republic.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3705328.3748087



This work is licensed under a Creative Commons Attribution 4.0 International License.

RecSys '25, Prague, Czech Republic

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1364-4/25/09
https://doi.org/10.1145/3705328.3748087

1 Introduction

Fairness is a complex and contested concept [23]. It is therefore not surprising that fairness in machine learning remains a challenging subject, especially in the area of recommender systems, where the presence of multiple stakeholders makes fairness inherently multifaceted [17]. Fairness concerns may appear on the *provider* side of the recommendation interaction; when the question is one of fairness among different items or individuals who provide these items to the system. Fairness may also arise on the *consumer* side of the interaction, the individual receiving recommendations from the system. There may also be others, e.g., the system operators or third-party stakeholders, for whom fairness is also of concern. Each of these stakeholders will have different, and sometimes competing, definitions and measurements of what they consider fair.

There is a burgeoning literature defining measures of fairness in recommender systems, studying the tradeoffs between fairness and measures of accuracy, and devising new algorithms with improved fairness properties [38]. What this literature largely shares are two key limitations as noted by Aird et al. [3], which hamper the practical adoption of these techniques:

- **Uni-dimensional Fairness:** Typically only a single dimension of fairness is considered, e.g., gender or location, when deriving fairness-aware recommender systems solutions. However, in real-world settings there may be multiple relevant dimensions along which fairness is important. It is often unclear how one-dimensional solutions can be extended to multiple dimensions.
- **Definitional Restrictions:** Typically only a single class of fairness definition is considered in given research work. Many proposed algorithmic solutions depend on the precise mathematical properties of a particular fairness definition and therefore cannot be applied in settings in which that

definition is not relevant. Research in applied settings has shown that a diversity of fairness definitions is needed to capture the range of concerns for recommendation [34, 35].

We follow the approach of Aird et al. [3] using the Social Choice for Recommendation Under Fairness - Dynamic (SCRUF-D) architecture, which instantiates fairness concerns as agents within the recommender system and uses social choice [8] to mediate between the fairness objectives and the recommender system itself. (More details on this framework can be found in Section 3.) This approach has the advantage of being flexible both in terms of the range of fairness definitions with which it is compatible as well as the number of different stakeholder concerns that can be simultaneously addressed. Prior work with SCRUF-D has shown that the technique is capable of supporting multiple heterogeneous provider-side fairness definitions [4]. However, research with SCRUF-D has so far concentrated only on provider-side fairness and only group fairness definitions. In this work we extend these considerations to include *individual fairness*, expanding the range of concerns and stakeholders that SCRUF-D is able to address.

Group fairness applies when concerns are associated with group membership: do members of group B receive worse treatment from the system than members of A? Group fairness relative to particular defined groups may be a statutory requirement under some legal regimes. For example, in the United States, subject to some exclusions, group fairness relative to the dimension of religion is a requirement in hiring, and this would extend to recommender systems being used in an employment context [6]. *Individual fairness* focuses on the fair treatment of individuals. The basic formula is that individuals who are highly similar, in all aspects relevant to the setting, should receive similar outcomes [16].

The research presented here goes beyond the previous work on provider-side group fairness in SCRUF-D to consider provider-side individual fairness. The distinction between individual and group fairness is often posed as a binary: some researchers assert that a given system can only pursue one or another [16]. However, applied research has shown that both types of concerns may be active in a given context [34, 35], and so combining individual and group fairness may be a practical necessity. The social choice formalism in SCRUF-D allows these concepts, with very different logics, to be simultaneously brought to bear within the same recommendation setting. In addition, since social choice aggregation procedures are formal and transparent (e.g., majority voting) they lend themselves to greater system transparency and explainability [18, 27].

We address the following research questions:

- **RQ1:** How can agents supporting provider-side individual fairness be implemented within SCRUF-D?
- **RQ2:** What is the interaction between fairness concerns, mechanisms for allocation and preference aggregation, and data set characteristics?
- **RQ3:** What is the interaction between group fairness and individual fairness objectives when both types are active?

Relative to these research questions, we show that provider-side individual fairness objectives can be implemented as fairness agents within SCRUF-D and serve to promote individual fairness objectives alongside group fairness ones. Code and data for our experiments along with our extensions of SCRUF-D are available. [1]

2 Related Work

Fairness in recommender systems is an active area of research surveyed in many recent publications [15, 17, 31, 38] and a number of open source tool-kits for evaluating [7] and building fair machine learning and/or recommendation systems [14]. As noted above, most research in this area assumes that only a single dimension of fairness is relevant in a given application and that the definition of fairness can be fixed in advance. Indeed, recent surveys and talks [14, 15, 34, 35] have noted that fairness is often too narrowly conceptualized, not addressing the full scope and spectrum of fairness concerns that real-world systems implementers require.

Several recent works have started to address this gap. There is existing research that recognizes the multiple stakeholders and multiple dimensions of fairness found in recommender systems. One research thrust emphasizes the interaction between fairness on the consumer and provider sides of the recommendation interaction [9, 30, 37, 40]. However, in all these works the definitions of fairness are fixed a prori, and act as a regularizer on the final outcomes. This formulation does not provide the flexibility of SCRUF-D which is able to handle a multitude of fairness definitions by design. In this work, we do not address multi-sided combinations of fairness, leaving that for future work. We note that there is no inherent limitation to such combinations in SCRUF-D, only that our present study focuses on the interactions between individual and group fairness definitions.

There are fewer contributions that allow for multiple fairness definitions on the provider side. There is prior work on the SCRUF-D architecture [4] on which this work builds, which shows that the architecture can support multiple heterogeneous group fairness definitions for providers. Another algorithm that goes beyond single fairness definitions is Multi-FR [41], which allows for multiple fairness concerns on both the provider and consumer side. However, Multi-FR is highly restrictive in its fairness definition, requiring mutually-exclusive provider groups where only one definition is applied to each group. This is not a general definition of provider-side group fairness; as shown in Smith et al. [34], fairness concerns in practice frequently have intersectional qualities.

In this work we formulate fairness as a temporal, dynamic concept, i.e., one that needs to be constantly re-evaluated as more recommendations are delivered by the system over time. Several works in this space, including those of Ge et al. [20] and Akpinar et al. [5] also investigate this problem, finding that a lack of attention to the temporal aspects of fairness, for example, by computing fair lists in batch mode before user arrivals, can lead to very unfair operation over time. However, these other works focus on single definitions of fairness, and do not consider the intersectional possibility of fairness over an evolving time horizon, i.e., different user arrival rates/types.

3 The SCRUF-D Architecture

Starting from the observations above – that fairness is an inherently contested concept [23] and that any fair recommender system

[1] https://github.com/that-recsys-lab/individual_fair_recsys2025.git

should be able to incorporate many different definitions of fairness [13] that are inherently multistakeholder [34] and multisided [10], Aird et al. [3, 4] introduced the Social Choice for Recommendation Under Fairness - Dynamic (SCRUF-D) platform as an extension to a simpler version first proposed by Sonboli et al. [36]. A general overview of the system can be seen in Figure 1 and we refer the reader to Aird et al. [3] for a complete description of the system. A full implementation of SCRUF-D is available on GitHub.[2]

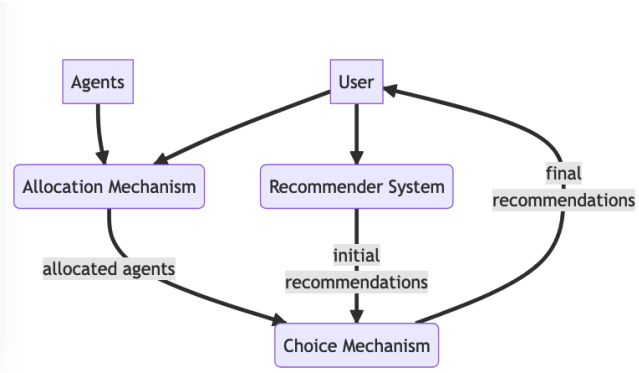


Figure 1: Overview of the SCRUF-D architecture

The SCRUF-D design starts from the observation that every user arrival is an opportunity to enhance the fairness of the overall system, in the context of the recommendations that have been delivered up to that point in time. In order to give voice to and balance between the different fairness concerns of the stakeholders of the system, including the user, the items, and the platform developer [1], we turn to ideas from social choice [8]. SCRUF-D embodies the fairness concerns of multiple stakeholders as a set of *fairness agents* that are *allocated* to a user when recommendations are being generated. Once the agents are allocated to a user, their ordering of items is aggregated with the user's preferences (here coming from a traditional recommendation algorithm) by using a *choice mechanism*, and the result is delivered to the user as their recommendation list.

Each fairness agent represents the concerns of one (set of) stakeholders of the system and is defined by three functions:

- a *fairness metric*, which is able to evaluate the history of the system and judge how fair the system has been to the agent over some time window,
- a *compatibility metric*, the agent's evaluation of how much they want to be matched to a particular user, and
- a *ranking function*, which expresses the agent's ordering/scoring of the items.

SCRUF-D does not make any assumptions about how the agents perform these functions, which provides great flexibility in defining and integrating different types of fairness.

Figure 1 provides an overview of what happens when a user arrives at the system to receive a recommendation. First, each of the fairness agents expresses their preference for being matched with, and intervening in the final recommendation to, the arriving user. The agents form this preference based on the user profile and the history of the recommendations delivered by the system so far. These preferences (scores) are then passed to an allocation mechanism which decides which agent or set of agents to assign to this user. This assignment may be a single agent or a weighting over all the agents, depending on the particular instantiation of the mechanism.

After agents are allocated, the system takes the preferences of the user, i.e., a scoring/ranking of items given by a traditional recommendation algorithm, and combines this preference with an scoring/ranking over items given by the allocated fairness agents. This gives us a set of scoring/ranking of items, one from the user, and others from the allocated fairness agents. These scoring/rankings are then combined into a final scoring/ranking from which the final recommendation is created using a social choice aggregation function, i.e., a voting rule. SCRUF-D can therefore be seen as reranker, operating over the original recommender system outputs.

The system evolves dynamically as the recommendations are issued to users. The fairness agents evaluate the recommendation lists delivered so far, and these judgments are used as part of the allocation mechanism for the next user. Fairness agents can be inserted or removed from the system without the need for retraining loss functions or other system designs. In this way, SCRUF-D is able to respond dynamically to the evolving system and fairness concerns.

3.1 Computational Complexity

Although the incorporation of the allocation and reranking stages within the SCRUF-D adds conceptual complexity to the recommendation process, the underlying computational complexity only adds a constant factor per recommendation list. Note that the reranking always takes place over a fixed set of items (in our experiments here that number is 50). The history window over which agents evaluate fairness is also fixed, so that agents need only consider interactions over a limited period, independent of the number of users. All of the social choice allocation and selection mechanisms we consider in this work are polynomial time (and in some cases linear time) computable in the number of agents and items [8, 43]. In our case the number of agents is bounded to three in our experiments, and the preferences are bounded by the fixed item set. Hence, the actual computational cost of these interventions is extremely low, and can be run in real-time at system deployment.

4 Fairness Agents

As described above, SCRUF-D integrates fairness into recommendation system as agents defined for each fairness concern. In this paper, we employ agents oriented towards both group and individual fairness on the provider side of the recommendation interaction.

4.1 Group Fairness

Group fairness concerns are those that seek to compensate for biases relative to particular groups, typically demographic groups. We focus on provider-side fairness and therefore we define agents that consider fairness relative to particular provider groups. There are many ways to define a recommender system's benefit or utility

[2]https://github.com/that-recsys-lab/scruf_d

for a group. Aird et al. [4] examines how SCRUF-D can integrate multiple different definitions of group fairness.

For our purposes, we focus on a single common definition for provider-side recommendation utility: recommendation proportion or exposure. That is, we measure the fraction of recommended items of some set of recommendation lists that are associated with a particular group. The more recommended items, the higher the utility.

Formally, let $\mathscr{L}$ be a list of recommendation slates delivered to some set of users: $[\ell_0, \ell_1, ..., \ell_j]$, where each ℓ is a list of items $[i_0, i_1, ..., i_k]$. Let $\mathbb{I}_g(i)$ be an indicator function that returns 1 if i is associated with group g. We can express exposure E as follows:

$$E_g = \frac{\sum_{\ell \in \mathscr{L}} \sum_{i \in \ell} \mathbb{I}_g(i)}{\sum_{\ell \in \mathscr{L}} |\ell|}$$

As described by Aird et al. [3], this value is normalized by a threshold value θ, defined as the degree of exposure considered to be fair in a given application setting: $\bar{E}_g = \min(1, E_g/\theta)$.

In our experiments, we set a somewhat-arbitrary threshold for the amount of exposure that is considered fair. In practice, this could be the proportion of in-group items in the catalog or some other value derived from stakeholder consultation [34]. The choice of dimensions along which to consider group fairness varies based on the datasets with which we are experimenting. See Section 5.3 for these details.

4.2 Individual Fairness

Individual fairness seeks fair outcomes measured at the individual level. The classic definition by Dwork et al. [16] holds that individuals who are similar, in all ways objectively relevant to the decision, should receive similar outcomes. This definition is difficult to apply to personalized recommendation as there is no objective source of item quality: some people genuinely desire recommendations that others would find bad. Even global measures like popularity or average rating may encode societal biases rather than some notion of universal objective quality.

In this work, we assume that the skewed item distributions produced by recommended systems, due to popularity bias and other factors, are inherently unfair and it is better to spread recommendations more widely across the item catalog [2]. While one may wish to ensure that notions of exposure [33], i.e., the expected number of times an item is recommended discounted by the position in those recommendations, is equalized, we opt for the conceptually and computationally simpler notion of coverage. Other models of provider utility we leave for future work.

Catalog coverage is a popular measure for the breadth and diversity of the recommendations delivered by a system [19, 24] and is commonly used as a measure of provider-side individual fairness [17]. Informally, coverage measures the proportion of items in the overall catalog that have been recommended to any user. Let $\mathscr{L}$ be a set of recommendation lists in a given time window and $\mathscr{C}$ be all unique items in the catalog.

$$Coverage(\mathscr{L}) = \frac{|\text{Unique Items in } \mathscr{L}|}{|\text{Total Items}|} = \frac{|\bigcup_{\ell \in \mathscr{L}} \ell|}{|\mathscr{C}|}$$

4.3 Compatibility

The purpose of the compatibility function is to determine the quality of match between a recommendation opportunity (user arrival) and a given fairness agent. Prior research has found that personalizing fairness in this way can allow for better tradeoffs between fairness and accuracy [26, 29]. Liu and Burke [26] use information entropy to measure the association between a user and the sensitive features of an item; we use a simpler proportional test: what fraction of liked items by the user contain the given feature? This quantity is normalized in two different ways: first, across all of the feature descriptors, relative to a given user, and then across all of the users, so the most compatible user has a score of 1 and the least compatible a score of zero.

To compute compatibility for our individual fairness agents, we counted how often items showed up in the ratings and normalized those counts with 0 meaning an item has been rated the most and 1 meaning an item has been rated the least. We then averaged those scores for items rated by each user to get their compatibility with the broader catalog of less popular items.

5 Methodology

5.1 Datasets

We use two different datasets for these experiments, the key details are summarized in Table 1. These datasets are:

- **MovieLens**: We use the MovieLens 1M dataset along with The Movies Dataset. The Movies Dataset contains features associated with the movies in the MovieLens 1M dataset that can be used for the assessment of group fairness [22].
- **Music**: We use the AMBAR dataset [21], which contains normalized play count information for music tracks. We binarized these ratings, considering all play counts as positive interactions. We selected random subsets of 6000 users on which to perform reranking due to the large number of users in this dataset.

Dataset	# Users	# Items	# Ratings
MovieLens	6,040	3,706	1,000,209
Music	31,013	443,921	3,311,462

Table 1: Dataset characteristics. For the Music dataset, samples of 6,000 users were extracted with associated items and ratings.

5.2 Recommendation Generation

Personalized recommendations are one of the key inputs to SCRUF-D. These are understood to represent the user's preferences in the preference aggregation stage. Any recommendation algorithm can be used, and prior work with the SCRUF-D architecture has employed a variety of algorithms [3, 4]. For the purposes of this experiment, we used the well-known Bayesian Personalized Ranking (BPR) algorithm for matrix factorization [32] as implemented in RecBole 2.0 [42]. We randomly partitioned the data into 80% training and 20% test sets, and used the training data to generate

lists of 50 recommendations for each user. For MovieLens, this partitioning was performed five times for cross-validation. For AMBAR, we extract five different samples of 6,000 users.

5.3 Agent Definitions

Burke et al. [11] emphasize that value-sensitive and participatory approaches are essential to devising evaluation measures in multi-stakeholder settings including recommendation fairness. Without stakeholder consultation along the lines of the studies in [34] and [35], it is not possible to have a principled determination of what would count as a fairness metric or appropriate target for any group. For the purposes of this work, we chose our targets for demonstration purposes to be "difficult" but not impossible given the recommendation baseline. Fairness definitions and targets are input parameters to the system, so they can be easily adjusted based on stakeholder consultation.

MovieLens. We implemented two group fairness agents for the MovieLens experiments. The *Women W/D* agent is designed to promote movies with women on the directing or writing team.[3] We defined this agent in terms of proportional fairness as defined in Section 4.1 and set the target proportion to 30%. The *NonEnglish* agent supports movies where primary language is not English. This is a small and under-recommended subset of titles in the dataset. Again using the proportional definition, we set the target to be 7%. For the individual fairness agent in this scenario, we applied Catalog Coverage as described in Section 4.2, setting the target at 0.85. Since the MovieLens data contains rating values, we removed items rated 1 or 2 when determining "liked items" for the purposes of computing compatibility. As noted above, the fairness targets are meant to be representative but not definitive. In a production environment, these targets would be set through domain- and application-specific analysis and stakeholder consultation.

Music. For the Music dataset, we implemented a single proportional fairness metric (*Women Artists*) for female artists similar to the director / writer agent for movies.[4] The target proportion was 25%. We treated the playcount-based rating feature as binary, reflecting user interest, and set the Catalog Coverage target to 0.65.

5.4 Mechanisms

Prior work with SCRUF-D found that different combinations of allocation and aggregation mechanisms result in a spectrum of trade-offs between fairness and accuracy. Following the example of Aird et al. [4], we apply examples of key mechanism types for the two social choice operations in SCRUF-D. Note that there are many other options in the social choice literature for allocation and choice and we expect to experiment with additional methods in future work [8].

For allocation, we consider mechanisms that deliver only a single agent into the choice phase: *Least Fair* and *Lottery*, and one mechanism that allocates all agents but with weights: *Weighted*. The Least Fair mechanism does not take user compatibility into account, simply picking the agent with the lowest normalized fairness, but some prior results found that it was competitive with more sophisticated allocation mechanisms under many settings. Following Aird et al. [4], the other two mechanisms we test compute a probability of allocating each agent as a function of fairness and compatibility. In the Lottery mechanism, these probabilities are used to non-deterministically chose one of the agents to allocate. In the Weighted mechanism, these probabilities become the weights for the allocation. Note that there are computational advantages to processing only a single fairness agent in the choice phase, so if the Lottery is competitive with the Weighted mechanism, it would be preferred. Prior research did not find a consistent advantage for the Weighted mechanism across datasets.

In the choice mechanism, we experiment with three different choice mechanisms employing different voting logics [43]. Note that in all cases we use a weighted method so that we can adjust the preference given to the recommender system outputs, i.e., giving more weight to the arriving users' preferences, enabling us to explore the balance between accuracy and fairness. *Borda* is a well-known voting method that assigns scores to each preference rank and sums the scores across voters [43]. The method contrasts with *Copeland*, which is a pair-wise method, using win-loss comparisons for each pair of items and scoring by the number of wins. Finally, as we have numerical scores from the recommendation algorithm, we use a numerical *Rescoring* method, that ranks items based on a linear combination of the preference from the agent(s) and the recommender. Prior work has found that the Rescoring method achieves a good balance between fairness and accuracy, likely because it can take advantage of the computed utilities coming from the recommender. Social choice methods typically assume that only ordinal ranks are available.

5.5 Evaluation

We evaluate each SCRUF-D experiment with multiple metrics. For summative evaluation of the accuracy of the final recommendation lists, we compute *NDCG*@10 using the held-out test data. We note that (as in all fairness-aware recommendation work), accuracy measures are likely underestimates because of observational bias in the data; users are less likely to have seen and rated the items for which fairness is sought.

To evaluate overall fairness outcomes relative to each fairness concern, we use the fairness logic inherent in the definition of the fairness agent. However, we evaluate fairness over the entire set of recommendations delivered, not just the look-back window that agents have access to during the experiment.

When plotting fairness versus accuracy, we use the $l^{1/2}$ norm over the fairness scores for each individual agent [28]. Let $\mathcal{F} = \{f_1, f_2, ..., f_k\}$ be the set of fairness agents with associated fairness scores $m_1, m_2, ..., m_k$. The norm is given by:

$$m_{1/2} = \frac{(\sum_{i \in \mathcal{F}} \sqrt{m_i})^2}{|\mathcal{F}|^2}. \quad (1)$$

When all of the m_i fairness scores are equal, the metric reduces to simply the mean, but when the scores differ, the norm will be below the mean because of the properties of the square root function. The divisor $|\mathcal{F}|^2$ is included to normalize the scores to be in the same

[3] Gender information (*male*, *female*, and *unknown*) is provided for cast and some production roles in the augmented *The Movies* dataset built on top of MovieLens 1M: https://www.kaggle.com/datasets/rounakbanik/the-movies-dataset.

[4] This dataset includes *male*, *female*, and *undefined* as values for the gender feature, where an undefined value may mean a mixed gender group.

0..1 range as the original fairness scores. This norm is preferable to the mean of the fairness scores because the mean may hide large differences across agents.

Another set of metrics that we compute are ones that provide insight into how the recommendation lists are changed through the reranking procedure. There are two metrics in this category *rank-biased overlap* (RBO) [39] and *normalized least item promoted* (NLIP), a metric for reranking that we are introducing in this paper.

RBO is a list similarity metric that puts higher weight on similarities at the top of the list, as appropriate in recommendation where the top items are likely to get the most attention, as opposed to edit distance or rank correlation, that treat all changes as the same regardless of where in the lists they occur. RBO also does not depend on the lists containing the same set of items or the same number of items, which is also appropriate for our reranking application.

Similar to the computation of mean average precision (MAP), RBO repeatedly computes the overlap of the sub-lists of the two lists to be compared beginning with at the top of the ranking. The top items are compared, then the top 2 items, and so on down the list. The proportion of overlap is averaged over all such lists. Because the top of the list is repeatedly incorporated in the calculation, differences at the top ranks will have a larger impact than those further down. The RBO score will be maximized when the lists are the same and is zero when there is no overlap.

Let ℓ_d be the sub-list of ℓ from position 1 to d, RBO is then given by:

$$RBO(\ell, \ell') = \frac{1}{k'} \sum_{d=1}^{k'} \frac{|\ell_d \cap \ell'_d|}{d} \tag{2}$$

where k' is the length of the shorter list.

Our new metric, NLIP, is designed to measure how deeply into the initial recommendation list that a reranker reaches to move an item into the final recommended list. The item that is lowest ranked (but not within the original recommended set) but still ends up in the reranked recommendation set is the *least promoted item*. If we return 10 items to the user, and the least item promoted has rank 12, we can say that the order of items coming from the recommender is not changed significantly. If the least item promoted has rank 50, that would count as a more significant change. This aspect of the reranking is not captured by RBO since that metric only examines the overlap in the top k' positions. We normalize item's rank by subtracting the length of the final reranked list k' and dividing by the length of the non-recommended portion of original recommendation list $k-k'$. If the recommendation list is unchanged by the reranking operation, NLIP is zero. Thus, the NLIP value is between 0 and 1 and can be compared across reranking experiments.

Let ℓ be the original base recommendation list with length k and let ℓ' be the final reranked list with length k'. Let $\rho(i, \ell)$ be the rank of item i in a list ℓ.

$$NLIP(\ell, \ell') = \frac{\max_{i \in \ell'} (\max(0, \rho(i, \ell) - k'))}{k - k'}$$

The intuition behind *NLIP* is similar to that of Kendall's τ measure that counts concordant and discordant pairs. The more that an item moves in the ranking, the more discordant pairs are created. *NLIP* has the specific aim of helping us understand the impact of the length of the original list on the fairness outcomes. If the *NLIP* value never reaches more than (for example) 0.75, then it may be possible to compute fewer recommendations in the generation step because the items at the end of the list are never shown to any user. If the value is close to 1.0, then it is possible that more initial recommendations are needed.

In our evaluations, we compare the system output with the original un-reranked output of the recommender. As noted above, there are no published recommendation algorithms that can incorporate multiple heterogeneous provider-side fairness objectives including both group and individual fairness.

5.6 Tuning Experiments

Each combination of allocation and choice mechanism was tested with varying weights on the recommender system to choose experiments with the greatest improvement in fairness while maintaining an acceptable loss in accuracy. We used a 15% nDCG cutoff based on results from [28] showing that up to a 17% loss in accuracy can be considered acceptable in media recommendation settings. Within the 15% cutoff, we chose the configuration with the highest fairness score to represent each mechanism. In the case of MovieLens, we found that fairness could be improved without NDCG loss and so this rule resulted in the choice of the mechanism with the best fairness.

6 Results

Tables 2 and 3 show the results for the best performing experiment from each dataset and fairness goal, averaged over the five data folds and with associated 95% confidence interval. We identify the mechanism combination with the best fairness norm given a 15% loss or less. Note that in these tables all fairness scores are the normalized values relative to fairness targets described in Section 5.3.

6.1 Individual fairness

In our first set of experiments, we added only the agent seeking individual fairness to study its properties in isolation and to see how SCRUF-D works with an individual fairness objective. In the Music dataset, we see little difference between choice mechanisms but Least Fair allocation boosts Coverage even above the desired target value. This comes at a cost to fairness for the Women Artists.

Our reranking metrics RBO and NLIP give a complementary picture of what is happening during reranking. For the Music dataset, RBO is low, around 0.1, meaning that there is very little carry-over between the original top ten items and the reranked output. Supporting this finding is the NLIP value of 0.830: in raw terms, this corresponds to an average rank of around 43 for the least promoted item in the original results list of size 50. We see that in order to achieve individual fairness, the reranking has to go deep into the recommendation list to find long-tail items.

For the MovieLens dataset, we see that recommendation accuracy in terms of nDCG actually improves after reranking for most conditions. This shows that personalization and fairness do not always trade-off, as has been observed in some prior work [12]. In this case, we attribute the phenomenon to our choice of the BPR algorithm, which is known for its popularity bias [25].

Condition	Allocation	Choice	nDCG	Coverage	Women	L1/2 Norm	RBO	NLIP
Baseline	–	–	0.103±0.002	0.562±0.004	0.832±0.009	0.690±0.004	–	–
Individual	Least Fair	*	0.095±0.002	1.209±0.004	0.613±0.011	0.886±0.007	0.115±0.002	0.830±0.002
Group	†	*	0.091±0.001	0.585±0.003	1.079±0.009	0.813±0.006	0.767±0.013	0.236±0.013
Both	Lottery	*	0.101±0.002	0.572±0.004	0.821±0.009	*0.691*±0.004	0.982±0.001	0.015±0.001

Table 2: Results for the Music dataset with 95% confidence interval from five-fold cross-validation. Highest fairness result was chosen with nDCG not less than 15% of baseline. * indicates results where there was no significant difference between mechanisms. †indicates results where there was no significance difference between Lottery and Weighted, but where Least Fair was not competitive. Italics indicate results not significantly different from the baseline. RBO or NLIP results are not calculated for the Baseline condition as no reranking is applied.

Condition	Allocation	Choice	nDCG	Coverage	Women W/D	Non-English	L1/2 Norm	RBO	NLIP
Baseline	–	–	0.073±0.002	0.661±0.056	0.821±0.029	0.363±0.022	0.599±0.022	–	–
Individual	Least Fair	Copeland	0.086±0.009	0.728±0.054	0.742±0.022	0.472±0.017	0.641±0.019	0.453±0.153	0.517±0.200
Group	Least Fair	Borda	*0.075*±0.008	*0.671*±0.055	1.000±0.000	1.000±0.000	0.883±0.021	0.745±0.189	0.288±0.161
Both	Lottery	Borda	0.085±0.008	0.728±0.053	0.878±0.009	0.496±0.019	0.691±0.023	0.435±0.145	0.536±0.187

Table 3: Results for the MovieLens dataset with 95% confidence interval from five-fold cross-validation. All results have better nDCG than the Baseline so the highest fairness result was chosen. Italics indicate results not significantly different from the baseline. RBO or NLIP results are not calculated for the Baseline condition as no reranking is applied.

We saw very similar accuracy / fairness results across mechanisms, with Least Fair / Copeland mechanism slightly outperforming the others. The Coverage increases only slightly, not nearly as much as in the Music dataset, and we see a mixed impact on the other fairness objectives: Women Writers and Directors do worse but non-English movies do better.

In MovieLens, the reranker does not have as difficult a task in meeting the individual fairness target. For the top mechanism, the RBO is 0.453, a smaller change in the top ten list. The NLIP score is below 0.517, which corresponds to a rank around 30. The reranker still, on average, has to go outside the top 20 items to increase individual coverage but not as far as in the Music dataset. This difference between the datasets' reranking behavior is expected since the Coverage of the original recommender is somewhat higher in MovieLens.

6.2 Group fairness

To provide a baseline for the joint optimization of group and individual fairness, we produced results for the group fairness objectives only. These are similar experiments to those that appear in prior work with SCRUF-D [3], but we are not attempting to reproduce these results exactly: we use different datasets and different base recommendations coming from RecBole.

In Music, all of the mechanisms have equivalent performance, except for Least Fair, which was not able to reach the 15% accuracy cut-off at any setting. Other mechanisms are similar to each other, easily meeting the fairness target and enhancing Coverage slightly at the same time. MovieLens has two group fairness agents and they are both able to reach the fairness target with the Least Fair / Borda mechanism combination without loss of accuracy or coverage.

The reranking metrics confirm at this tradeoff is due to a relatively small amount of reranking being imposed: RBO values are relatively high and the NLIP value are similar. In MovieLens, the value is slightly higher because it is needing to meet the needs of two fairness objectives at once.

6.3 Combining individual and group fairness

Finally, we experiment with combined optimization of group and individual fairness. This is the *Both* condition in the tables. In MovieLens, SCRUF-D is able to support enhanced fairness even when we have multiple agents with very different fairness logics. We note that this is the most difficult condition for the system as it is balancing (in the MovieLens case) three different fairness concerns at once. Especially striking is the fact that the MovieLens dataset still shows increased accuracy.

The Music dataset is different. The system struggles to achieve both of these objectives at the same time, with Coverage increasingly slightly and Women Artists have exposure that is not significantly different from the Baseline. This is perhaps not surprising given that these objectives are countervailing in the Individual condition. It is also striking that ultimately little reranking is employed with NLIP less than 0.02: on average less than one promoted item per recommendation list.

The Least Fair mechanism, which was excluded from the table due to the 15% accuracy cutoff, does a much better job of enhancing Women Artists, but it does not improve Coverage. We expect this is because the Women Artist objective is hard to meet and therefore the associated agent has low fairness and is allocated most of the time, leaving little space for the Coverage agent to impact the rankings.

These tradeoffs for the best performing configurations can also be seen in Figures 2 and 3. In the Music data, we see a big improvement in the individual fairness metric when only its associated agent is allocated, as expected, and group fairness here is below the baseline. The inverse happens when only the group fairness agent is allocated, but here the accuracy takes a bigger hit. When

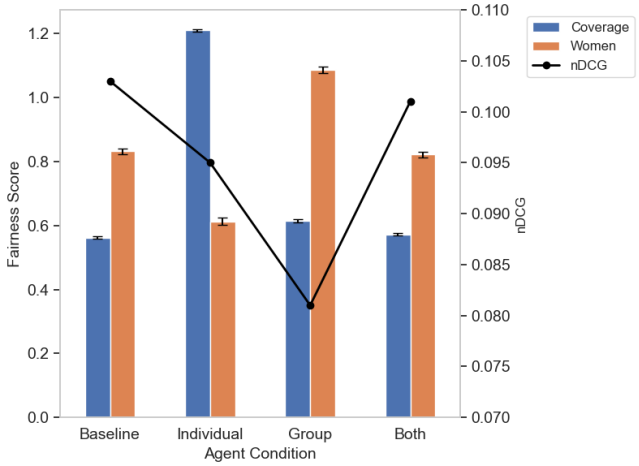


Figure 2: Accuracy and Agent-level Fairness: Music Dataset

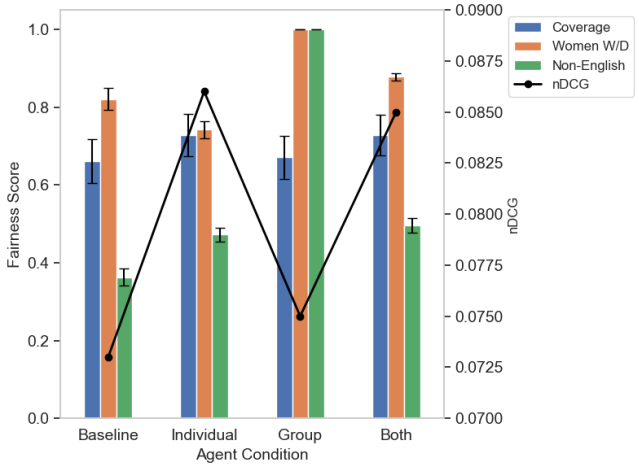


Figure 3: Accuracy and Agent-level Fairness: Movie Dataset

all agents are allocated, the result is not that different from the baseline.

The Movie experiments show a different pattern. Coverage only improves slightly across conditions, no doubt because it is already high. In the All agents condition, lower provider fairness is seen, particularly for the Non-English group. The Non-English objective is boosted by the individual fairness agent and the Women W/D group is not. We discuss this phenomenon more below.

In Figures 4 and 5, we provide more detailed results for the experiments across the different mechanism choices when all agents are active. Figure 4 plots NDCG versus the $m_{1/2}$ norm for the three agents in the MovieLens experiment. Note that the Baseline recommender would far below and to the left. The three Lottery conditions are far better in terms of fairness, while the Weighted condition (which in theory should be similar to Lottery in expectation) does not provide as much fairness. We believe this is because in the multiple agent case the recommender can outvote the agents when they disagree with each other, and in the Lottery case, only a single agent is present to push its fairness preferences.

The Weighted mechanism occupies a different position on the Pareto frontier with higher accuracy at a lower degree of fairness improvement. This is not selected as our top mechanism because our criterion was not accuracy gain but the best fairness at the least loss.

Interestingly, the Least Fair allocation, which in prior work was found to be dominant in some cases and works well on this dataset in the Individual condition, is behind in terms of the fairness / accuracy tradeoff. Further analysis is required but it could be that the some agent or agents is being "starved" and left unallocated when user compatibility is ignored.

The picture changes in Figure 5 with the Music data. We get two clusters of mechanisms in the upper right (dominant position) and lower left. Note that Least Fair does not appear at all because it did not meet the accuracy cutoff in this dataset. The Weighted mechanism appears to do well, but we should not read too much into these differences as they are all below the threshold of significance.

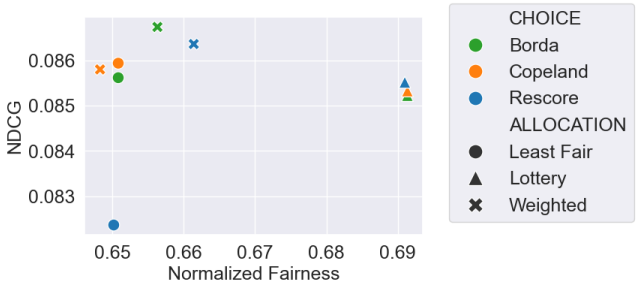


Figure 4: Accuracy vs Normalized Fairness across Mechanisms: MovieLens Dataset, Combined Individual and Group Fairness

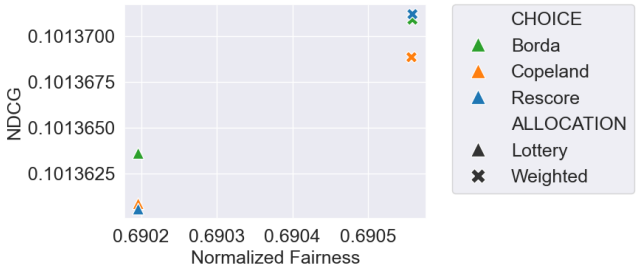


Figure 5: Accuracy vs Normalized Fairness across Mechanisms: Music Dataset, Combined Individual and Group Fairness. Least Fair mechanism is excluded

6.4 Correlation Between Fairness Scores

These results show that simultaneous individual and group fairness objectives can be addressed through a social choice approach, but the success of the method is domain- and objective-dependent. We explore one domain-specific here, which is the relationship between

individual and group fairness. We examined correlations among the fairness scores over the different agents across the mechanisms studied. Given the dichotomy often expressed between group and individual fairness, for example in [16], we might expect to see these quantities anti-correlated. On the other hand, protected group items are, almost by definition, underrepresented in recommendation outputs so individual fairness should improve their prospects. To understand this phenomenon, we calculated the correlation between these types of fairness across all of our experimental conditions.

Evidence for the antagonistic effect in the Music dataset can be seen in the tension between individual fairness and our group fairness concern (Women Artists). These are moderately anti-correlated with a Pearson coefficient of -0.534. One way to understand this effect is to consider the existence of a small number of very popular women musical artists (consider, for example, Taylor Swift and Beyoncé). By modestly promoting these artists' music, the system can improve the fairness metric for their associated group. Because these artists are already popular, there is not a big accuracy cost in doing this. On the other hand, this minor reranking of what are already popular "short head" songs does not do much for long-tail items and their creators. So, the individual and group agents are really trying to accomplish different things.

The picture is more mixed when we look at MovieLens. Because there are three agents, we compute the correlation matrix shown in Table 4. The Women Writer / Director agent operates similarly to what we see in the Music data. It is somewhat anti-correlated with individual fairness (-0.416), probably for similar reasons: there are women directors making very popular movies: Greta Gerwig's *Barbie* (2023) comes to mind. On the other hand, the Non-English fairness concern is more mildly anti-correlated with individual fairness. MovieLens, having a largely US user base, reflects American taste for English-language movies so there aren't non-English blockbusters in the same way as there are popular female directors and writers. There are not as many short-head items to promote and so promoting these non-English movies does not decrease Coverage as much. Interestingly, the two group fairness metrics are highly correlated. Possibly this is because female directors are more common among the non-English pool of movies. More investigation is required to understand this effect.

	Coverage	Women W/D	Non-English
Coverage	1.000	–	–
Women W/D	-0.416	1.000	–
Non-English	-0.210	0.702	1.000

Table 4: Fairness outcomes correlation (MovieLens experiments)

7 Conclusion

In this paper, we have shown that the SCRUF-D architecture allows for the combination of individual and group provider-side fairness in recommendation. This architecture instantiates of fairness objectives as agents, which participate independently in the social choice operations of allocation and preference allocation that lead to reranked recommendations. Prior work showed that SCRUF-D can combine multiple heterogeneous group fairness definitions. We extend that work with the addition of individual fairness in the form of individual fairness. Across multiple datasets, we show individual fairness and group fairness can be simultaneously improved, but there are tensions between them that the system must manage.

Addressing RQ1, we have shown that provider-side individual fairness agents can be implemented within SCRUF-D using the same mechanisms and designs used to define group fairness agents in prior work. We chose catalog coverage as measure of individual fairness but other definitions could be employed (for example, Gini index) and we expect to consider these in future work.

We did not find consistent patterns of interactions between mechanisms for allocation and preference aggregation across datasets and so it is difficult to have a definitive answer to RQ2. In the MovieLens dataset, we saw the Lottery allocation mechanism with the Borda preference aggregation outperform all other mechanisms in terms of fairness compared to other mechanisms. We did not see many clearly dominant mechanism combinations with the Music dataset, and the system did not greatly enhance individual and group fairness when all agents were active.

Our findings for RQ3 were positive: individual fairness and group fairness can improve simultaneously depending on the dataset, but individual fairness is not always enough to guarantee group fairness. The impact individual fairness has on group fairness depends on the group fairness definition and the characteristics of the protected group. Individual fairness can have either a positive or a negative correlation with group fairness. We saw a steeper tradeoff for group fairness versus individual fairness in the Music data where there was an anti-correlation between the two objectives.

Another contribution of this work is the Normalized Least Item Promoted (NLIP) metric. We show the value of this metric in tandem with a list comparison metric like RBO in understanding the inner workings of a reranker. We believe NLIP can also be useful in tuning reranking pipelines, for example, to avoid generating unnecessarily long base recommendation lists, if items at the ends of these lists would remain untouched by subsequent operations.

In future work, we intend to explore additional definitions of individual fairness on the provider side, including Gini index, expected exposure, and others. We also intend to look at utility on a provider-by-provider basis, rather than the item-by-item basis implemented here. We also plan to extend SCRUF-D to encompass consumer-side fairness.

Acknowledgments

Authors Burke and Aird were supported by the National Science Foundation under grant awards IIS-1911025 and IIS-2107577. Nicholas Mattei was supported by NSF Grant IIS-2107505. Author Stefancova was supported by national grants APVV-20-0353 and APVV-23-0292.

References

[1] Himan Abdollahpouri, Gediminas Adomavicius, Robin Burke, Ido Guy, Dietmar Jannach, Toshihiro Kamishima, Jan Krasnodebski, and Luiz Pizzato. 2020. Multistakeholder recommendation: Survey and research directions. *User Modeling and User-Adapted Interaction* 30 (2020), 127–158.

[2] Himan Abdollahpouri, Masoud Mansoury, Robin Burke, and Bamshad Mobasher. 2020. The connection between popularity bias, calibration, and fairness in recommendation. In *Proceedings of the 14th ACM conference on recommender systems*. 726–731.

[3] Amanda Aird, Paresha Farastu, Joshua Sun, Elena Stefancová, Cassidy All, Amy Voida, Nicholas Mattei, and Robin Burke. 2024. Dynamic fairness-aware recommendation through multi-agent social choice. *ACM Transactions on Recommender Systems* 3, 2 (2024), 1–35.

[4] Amanda Aird, Elena Štefancová, Cassidy All, Amy Voida, Martin Homola, Nicholas Mattei, and Robin Burke. 2024. Social Choice for Heterogeneous Fairness in Recommendation. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 1096–1101.

[5] Nil-Jana Akpinar, Cyrus DiCiccio, Preetam Nandy, and Kinjal Basu. 2022. Long-term Dynamics of Fairness Intervention in Connection Recommender Systems. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*. ACM, New York, NY, USA, 22–35.

[6] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2017. *Fairness in machine learning*. https://fairmlbook.org/pdf/fairmlbook.pdf

[7] Rachel KE Bellamy, Kuntal Dey, Michael Hind, Samuel C Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilović, et al. 2019. AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. *IBM Journal of Research and Development* 63, 4/5 (2019), 4–1.

[8] F. Brandt, V. Conitzer, U. Endriss, J. Lang, and A. D. Procaccia (Eds.). 2016. *Handbook of Computational Social Choice*. Cambridge University Press.

[9] Francois Buet-Golfouse and Islam Utyagulov. 2022. Towards fair multi-stakeholder recommender systems. In *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization*. ACM, New York, NY, USA, 255–265.

[10] Robin Burke. 2017. Multisided Fairness for Recommendation. , 5 pages. arXiv:1707.00093 [cs.CY] https://arxiv.org/abs/1707.00093 Presented at Workshop on Fairness, Accountability and Transparency in Machine Learning (FATML).

[11] Robin Burke, Gediminas Adomavicius, Toine Bogers, Tommaso Di Noia, Dominik Kowald, Julia Neidhardt, Özlem Özgöbek, Maria Soledad Pera, Nava Tintarev, and Jürgen Ziegler. to appear. De-centering the (Traditional) User: Multistakeholder Evaluation of Recommender Systems. *ACM Transactions on Recommender Systems* (to appear).

[12] Robin Burke, Nasim Sonboli, Masoud Mansoury, and Aldo Ordoñez-Gauger. 2017. Balanced Neighborhoods for Fairness-aware Collaborative Recommendation. In *Workshop on Responsible Recommendation (FATRec)*.

[13] Robin Burke, Amy Voida, Nicholas Mattei, Nasim Sonboli, and Farzad Eskandanian. 2022. Algorithmic fairness, institutional logics, and social choice. *Social Responsibility of Algorithms 2022* (2022), 5 pages.

[14] Henriette Cramer, Kenneth Holstein, Jennifer W. Vaughan, Hal Daumé III, Miroslav Dudík, Hanna Wallach, Sravana Reddy, and Jean Garcia-Gathright. 2019. Challenges of incorporating algorithmic fairness into industry practice.

[15] Yashar Deldjoo, Dietmar Jannach, Alejandro Bellogin, Alessandro Difonzo, and Dario Zanzonelli. 2024. Fairness in recommender systems: research landscape and future directions. *User Modeling and User-Adapted Interaction* 34, 1 (2024), 59–108.

[16] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. 2012. Fairness through awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*. ACM, 214–226.

[17] Michael D Ekstrand, Anubrata Das, Robin Burke, Fernando Diaz, et al. 2022. Fairness in information access systems. *Foundations and Trends® in Information Retrieval* 16, 1-2 (2022), 1–177.

[18] Rachel Freedman, Jana Schaich Borg, Walter Sinnott-Armstrong, John P Dickerson, and Vincent Conitzer. 2020. Adapting a kidney exchange algorithm to align with human values. *Artificial Intelligence* 283 (2020), 103261.

[19] Mouzhi Ge, Carla Delgado-Battenfeld, and Dietmar Jannach. 2010. Beyond accuracy: evaluating recommender systems by coverage and serendipity. In *Proceedings of the fourth ACM conference on Recommender systems*. 257–260.

[20] Yingqiang Ge, Shuchang Liu, Ruoyuan Gao, Yikun Xian, Yunqi Li, Xiangyu Zhao, Changhua Pei, Fei Sun, Junfeng Ge, Wenwu Ou, et al. 2021. Towards long-term fairness in recommendation. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*. ACM, New York, 445–453.

[21] Elizabeth Gómez, David Contreras, Ludovico Boratto, and Maria Salamo. 2024. AMBAR: A dataset for Assessing Multiple Beyond-Accuracy Recommenders. In *Proceedings of the 18th ACM Conference on Recommender Systems*. 137–147.

[22] F Maxwell Harper and Joseph A Konstan. 2015. The MovieLens Datasets: History and Context. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 5, 4 (2015), 19.

[23] Abigail Z Jacobs and Hanna Wallach. 2021. Measurement and fairness. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. 375–385.

[24] Aryan Jadon and Avinash Patil. 2024. A comprehensive survey of evaluation techniques for recommendation systems. In *International Conference on Computation of Artificial Intelligence & Machine Learning*. Springer, 281–304.

[25] Dietmar Jannach, Lukas Lerche, Iman Kamehkhosh, and Michael Jugovac. 2015. What recommenders recommend: an analysis of recommendation biases and possible countermeasures. *User Modeling and User-Adapted Interaction* 25 (2015), 427–491.

[26] Weiwen Liu and Robin Burke. 2018. Personalizing Fairness-aware Re-ranking. arXiv:1809.02921 [cs.IR] https://arxiv.org/abs/1809.02921 Presented at the 2nd FATRec Workshop held at RecSys 2018, Vancouver, CA..

[27] Nicholas Mattei. 2020. Closing the Loop: Bringing Humans into Empirical Computational Social Choice and Preference Reasoning. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI 2020*. 5169–5173.

[28] Rishabh Mehrotra, James McInerney, Hugues Bouchard, Mounia Lalmas, and Fernando Diaz. 2018. Towards a Fair Marketplace: Counterfactual Evaluation of the Trade-off between Relevance, Fairness & Satisfaction in Recommendation Systems. In *Proceedings of the Conference on Information and Knowledge Management*. 2243–2251.

[29] Mohammadmehdi Naghiaei, Hossein A Rahmani, and Yashar Deldjoo. 2022. Cpfair: Personalized consumer and producer fairness re-ranking for recommender systems. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 770–779.

[30] Gourab K Patro, Arpita Biswas, Niloy Ganguly, Krishna P Gummadi, and Abhijnan Chakraborty. 2020. FairRec: Two-Sided Fairness for Personalized Recommendations in Two-Sided Platforms. In *Proceedings of The Web Conference 2020*. ACM, New York, NY, USA, 1194–1204.

[31] Gourab K Patro, Lorenzo Porcaro, Laura Mitchell, Qiuyue Zhang, Meike Zehlike, and Nikhil Garg. 2022. Fair ranking: a critical review, challenges, and future directions. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, USA, 1929–1942.

[32] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2009. BPR: Bayesian personalized ranking from implicit feedback. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence*. 452–461.

[33] Ashudeep Singh and Thorsten Joachims. 2018. Fairness of exposure in rankings. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, 2219–2228.

[34] Jessie J Smith, Anas Buhayh, Anushka Kathait, Pradeep Ragothaman, Nicholas Mattei, Robin Burke, and Amy Voida. 2023. The Many Faces of Fairness: Exploring the Institutional Logics of Multistakeholder Microlending Recommendation. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, USA, 1652–1663.

[35] Jessie J Smith, Aishwarya Satwani, Robin Burke, and Casey Fiesler. 2024. Recommend Me? Designing Fairness Metrics with Providers. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*. 2389–2399.

[36] Nasim Sonboli, Robin Burke, Nicholas Mattei, Farzad Eskandanian, and Tian Gao. 2020. "And the Winner Is...": Dynamic Lotteries for Multi-group Fairness-Aware Recommendation. arXiv:2009.02590 [cs.IR]

[37] Tom Sühr, Asia J Biega, Meike Zehlike, Krishna P Gummadi, and Abhijnan Chakraborty. 2019. Two-sided fairness for repeated matchings in two-sided markets: A case study of a ride-hailing platform. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, New York, NY, USA, 3082–3092.

[38] Yifan Wang, Weizhi Ma, Min Zhang, Yiqun Liu, and Shaoping Ma. 2023. A survey on the fairness of recommender systems. *ACM Transactions on Information Systems* 41, 3 (2023), 1–43.

[39] William Webber, Alistair Moffat, and Justin Zobel. 2010. A similarity measure for indefinite rankings. *ACM Transactions on Information Systems (TOIS)* 28, 4 (2010), 1–38.

[40] Guoli Wu, Zhiyong Feng, Shizhan Chen, Hongyue Wu, Xiao Xue, Jianmao Xiao, Guodong Fan, Hongqi Chen, and Jingyu Li. 2024. FairSort: Learning to Fair Rank for Personalized Recommendations in Two-Sided Platforms. *IEEE Transactions on Knowledge and Data Engineering* (2024), 641–654.

[41] Haolun Wu, Chen Ma, Bhaskar Mitra, Fernando Diaz, and Xue Liu. 2022. A multi-objective optimization framework for multi-stakeholder fairness-aware recommendation. *ACM Transactions on Information Systems* 41, 2 (2022), 1–29.

[42] Wayne Xin Zhao, Yupeng Hou, Xingyu Pan, Chen Yang, Zeyu Zhang, Zihan Lin, Jingsen Zhang, Shuqing Bian, Jiakai Tang, Wenqi Sun, et al. 2022. RecBole 2.0: towards a more up-to-date recommendation library. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 4722–4726.

[43] William S. Zwicker. 2016. Introduction to the Theory of Voting. In *Handbook of Computational Social Choice*, Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia (Eds.). Cambridge University Press, 23–56. https://doi.org/10.1017/CBO9781107446984.003